

ASSIGNMENT 2

1. Write a Python code to perform Breadth First Search traversal on a graph

```
In [2]: from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    traversal = []

    while queue:
        node = queue.popleft()
        if node not in visited:
            traversal.append(node)
            visited.add(node)
            queue.extend(graph[node])
    return traversal

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': [],
}
bfs(graph, 'A')
```

```
Out[2]: ['A', 'B', 'C', 'D', 'E', 'F']
```

1. Write a Python code to perform Depth First Search traversal on a graph

```
In [3]: def dfs(graph, start, visited=None):
    if visited is None:
        visited = set()
        traversal = []

    if start not in visited:
        traversal.append(start)
        visited.add(start)
        for neighbor in graph[start]:
            traversal.extend(dfs(graph, neighbor, visited))
    return traversal

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': [],
}
dfs(graph, 'A')
```

```
Out[3]: ['A', 'B', 'D', 'E', 'F', 'C']
```